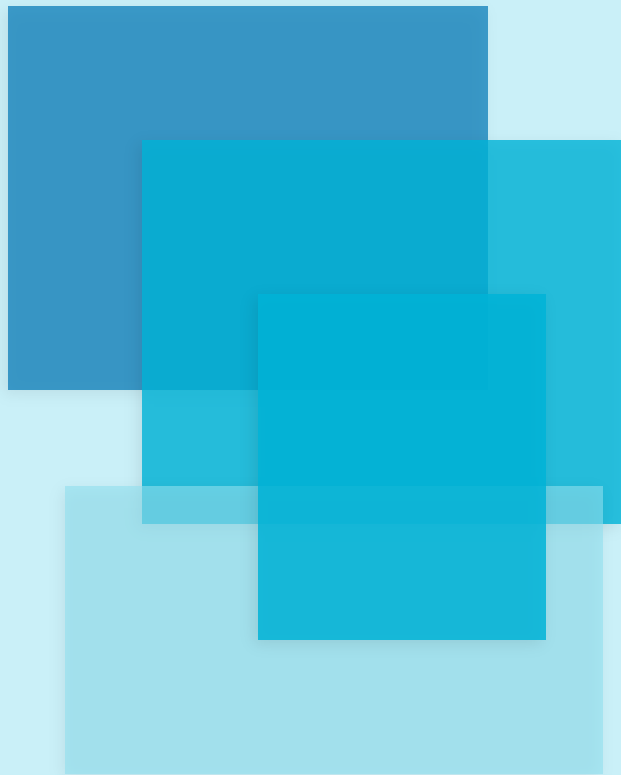

Are You Getting What You Pay For?

Auditing Model Substitution in LLM
APIs

Why Software Tests Fail and Hardware Attestation Wins

arxiv.org/abs/2504.04715



The Model Substitution Problem

Economic Incentives Drive Covert Swaps

1 Smaller Model

8B served instead of 70B; dramatic cost reduction with obvious quality loss

2 Quantized Variant

FP8/INT8 instead of FP16; minimal benchmark impact, hard to detect

3 Fine-Tuned Version

Modified weights or updated checkpoint; distribution shift undetectable without reference

4 Different Family

Entirely different architecture; requires only output-level similarity

H_0 : API serves $P_{spec}(y|x)$ — the specified model

H_1 : API serves $P_{actual}(y|x)$ — an alternative model

Auditor must distinguish these distributions from output samples alone

FP8 Quantization Preserves Most Accuracy

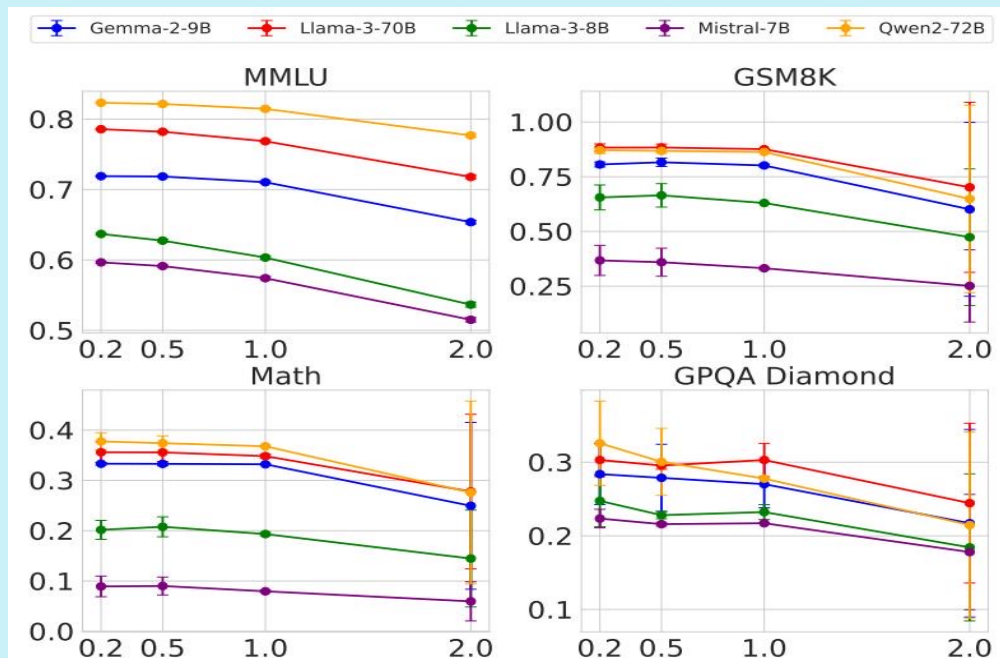
Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

FP8 scores fall within error bars of original — especially on MMLU and GSM8K. Detection via benchmarks alone is unreliable.

Source: arxiv.org/abs/2504.04715

Higher Temperature Amplifies Performance Gaps but Increases Noise



Source: arxiv.org/abs/2504.04715, Figure 1

Temperature scaling from 0.2 to 2.0 degrades performance across all five models and four tasks

At higher temperatures, the gap between original and substitute models narrows or becomes noisy

Statistical detection becomes harder as signal-to-noise ratio drops with increasing temperature

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Comparing token-level log probabilities between a local reference and the API seems promising — but production environments introduce significant nondeterminism.

- Software: transformers 4.40 vs 4.50
- Software: vLLM 0.6.2 vs 0.8.2
- Hardware: A100 vs H100 GPUs

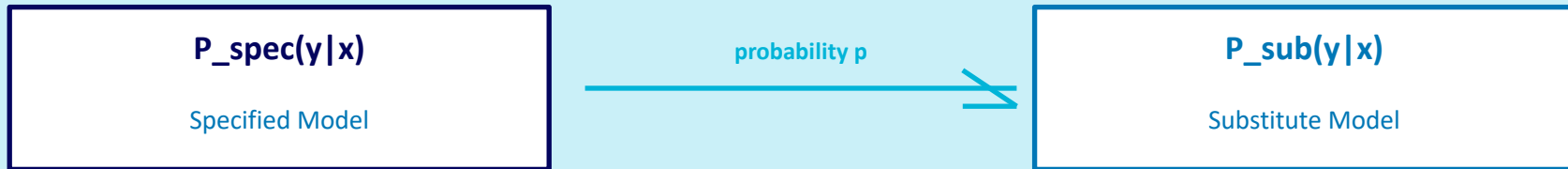
An auditor cannot distinguish legitimate infrastructure variation from actual model substitution.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Source: arxiv.org/abs/2504.04715, Figure 5

Randomized Routing at Low Rates

Makes Statistical Detection Query-Prohibitive



$$P_{\text{mixed}} = (1 - p) \cdot P_{\text{spec}} + p \cdot P_{\text{sub}}$$

$p = 0.5$

50% substitute queries; detectable with moderate query budget

$p = 0.1$

10% substitute queries; requires 10x more queries for same detection power

$p \rightarrow 0$

Mixed distribution converges to P_{spec} ; detection becomes exponentially harder

As $p \rightarrow 0$, the mixed distribution is indistinguishable from the specified model. Auditing requires $O(1/p^2)$ queries — becoming query-prohibitive for low substitution rates.

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Core Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Requires impractically large query budgets; fails against subtle substitutions	Cannot distinguish FP8 from original within error bars
Log-Probability Methods	Defeated by inference nondeterminism across software versions and hardware	Framework/hardware variation exceeds substitution signal
Adversarial Countermeasures	Benchmark evasion routes known prompts to genuine model; randomized routing dilutes signal	Detection cost grows as $O(1/p^2)$ — query-prohibitive at low p

No software-only method can provide reliable detection under realistic adversarial conditions.

Key insight: Each method fails independently — combining them does not yield a reliable composite detector. The fundamental issue is that the space of legitimate model outputs is too large and too variable to distinguish from substitution artifacts.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Software Methods

-  Statistical tests: query-prohibitive
-  Log-prob fingerprinting: defeated by nondeterminism
-  Adversarial evasion: fundamentally unaddressable

UNRELIABLE

TEE-Based Attestation

-  Provable: cryptographic attestation of model weights
-  Efficient: modest performance overhead only
-  Robust: immune to software-level evasion

PROVABLY SECURE

NVIDIA Confidential Computing on H100 GPUs provides practical TEE deployment today

Unlike ZKPs, TEEs are computationally feasible for models with billions of parameters

Key Takeaways

Close the Verification Gap with Hardware Attestation

1

Model substitution is economically motivated and practically undetectable via software — providers save GPU costs while maintaining pricing

2

FP8 quantization barely affects benchmark scores — performance drops fall within error bars on MMLU and GSM8K

3

Inference nondeterminism across frameworks and hardware kills log-probability fingerprinting as a detection method

4

Randomized substitution at low rates makes statistical detection query-prohibitive — requiring $O(1/p^2)$ queries

5

Trusted Execution Environments are the only currently viable path to verifiable LLM APIs — provable, efficient, and deployable today

The community must adopt hardware-based attestation standards to ensure LLM API integrity.